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Core 90 - Dynamic programming and memoization

9 questions to study here, each on its own page, with the full interview answer and code

Nothing to skip: every question that fits this topic is covered here

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MEDIUM

How to use

- Questions run from Easy to Medium to Hard, newest last-asked date first inside each level.
- PDFs run from the one with the most questions to the fewest. A question has pages in only one PDF; the others list it as SKIP
- Every page has < Prev | Contents | Next > at the bottom. Tap a title in the Contents to jump to it.
- Select code and press Cmd/Ctrl+C to copy it.

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STUDY PATH: question 23 of 167 (Core 90)

Priority score 7.46 | Dynamic programming and memoization | asked 21 days ago | frequency Single report

MEDIUM | 1Point3Acres | phone-screen | 1 of 9

Grid Path Count with Diagonal Moves and Checkpoints

ASKED 2026-09-29 | 21 DAYS AGO | LAST 30 DAYS

STUDY: TYPE	ALL TIME 48.9	30 DAYS 33	PRIORITY 35/228

ABOUT THIS QUESTION

Topic: Core 90 - Dynamic programming and memoization

Optimal answer from: written and tested here for this exact question

Related in your app (a neighbouring problem, not this exact question): LeetCode #62 Unique Paths

LeetCode match: LC 62 Unique Paths (related)

Frequency: Single report | Last asked: 2026-09-29 | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/grid-path-count-three-moves-checkpoints

QUESTION

Count paths in an n x m matrix from (n-1, 0) to (n-1, m-1) moving up-right, right, or down-right. Follow-ups: count paths visiting all checkpoints; then checkpoints must be visited in order, with the observation that each move changes column, and optimize space.

Matrix of size n x m. Start at (n-1, 0), end at (n-1, m-1). Allowed moves from (r, c): up-right (r-1, c+1), right (r, c+1), down-right (r+1, c+1); stay inside the matrix.
Return the total number of distinct paths from start to end.
Follow-up 1: the matrix contains checkpoints; count paths that visit all checkpoints.
Follow-up 2: the checkpoints must be visited in order. Since every move changes the column, each column can contain at most one visited checkpoint. Count the paths and optimize space.

APPROACH

Write the column-by-column DP where dp[c][r] depends only on dp[c-1][r-1..r+1], then reduce it to a single rolling column.
For checkpoints, practice the "segment" approach: sort by column, and between consecutive checkpoints only allow paths that pass through them (zero out other rows in a checkpoint's column).
Test: n = 1 (only straight right), m = 1, a checkpoint at the start or end, and two checkpoints in the same column.

HOW TO THINK ABOUT IT

Every move goes exactly one column to the right, so a path visits each column exactly once and the state is just 'which row am I in at this column'. That is a DP over columns with a single rolling array. A checkpoint (r, c) simply says the path must be in row r at column c, so you zero every other row of that column.

Basis: LeetCode 62, Unique Paths, with up-right / right / down-right moves, and checkpoints the path must visit.

BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def count(r, c, seen):
    if c == m - 1:
        done = r == n - 1 and checkpoints <= seen
        return 1 if done else 0
    total = 0
    for dr in (-1, 0, 1):
        nr = r + dr
        if 0 <= nr < n:
            nxt = seen | {(nr, c + 1)}
            total += count(nr, c + 1, nxt)
    return total
answer = count(n - 1, 0, {(n - 1, 0)})
```

TIME	SPACE
O(3^m)	O(m)

WHY IT IS SLOW

Up to 3^m paths. Every move advances one column, so a dp over (column, row) with a rolling array counts them in O(n * m).

OPTIMAL SOLUTION · Column DP with a rolling array and checkpoint filters

TIME	SPACE
O(n * m)	O(n)

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 Two checkpoints share a column.

No path can visit both, because a path is in exactly one row per column, so the answer is 0 (the code returns it before the DP). Checkpoints at the start or end cell are fine: they only filter rows of the first or last column.

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# MAIN QUESTION: Grid Path Count with Diagonal Moves and
# Checkpoints
#
```

```python
# Count paths in an n x m matrix from (n-1, 0) to (n-1,
# m-1) moving up-right, right, or down-right. Follow-ups:
# count paths visiting all checkpoints; then checkpoints
# must be visited in order, with the observation that each
# move changes column, and optimize space. Matrix of size
# n x m. Start at (n-1, 0), end at (n-1, m-1). Allowed
# moves from (r, c): up-right (r-1, c+1), right (r, c+1),
# down-right (r+1, c+1); stay inside the matrix. Return
# the total number of distinct paths from start to end.
# Follow-up 1: the matrix contains checkpoints; count
# paths that visit all checkpoints. Follow-up 2: the
# checkpoints must be visited in order. Since every move
# changes the column, each column can contain at most one
# visited checkpoint. Count the paths and optimize space.
#
# FOLLOW-UPS (answered below, in this order):
# 1. Two checkpoints share a column.
#
# INTERVIEW NOTES
# Every move goes exactly one column to the right, so a
# path visits each column exactly once and the state is
# just 'which row am I in at this column'. That is a DP
# over columns with a single rolling array. A checkpoint
# (r, c) simply says the path must be in row r at column
# c, so you zero every other row of that column.
#
# BRUTE FORCE (say it first): Enumerate every path
# Depth-first search over all move sequences, counting
# those that end at the bottom-right with the checkpoints
# visited. Exponential in the number of columns. Time
# O(3^m). Space O(m).


# BRUTE FORCE (say it first): Enumerate every path | time
# O(3^m)
def count_paths_brute(
    n, m, checkpoints=(), ordered=False
):
    cps = [tuple(c) for c in checkpoints]
    total = 0

    def go(r, c, seen):
```

```python
        nonlocal total
        if (r, c) in cps:
            seen = seen + [(r, c)]
        if c == m - 1:
            if r == n - 1 and (
                seen == cps
                if ordered
                else sorted(seen) == sorted(cps)
            ):
                total += 1
            return
        for dr in (-1, 0, 1):
            if 0 <= r + dr < n:
                go(r + dr, c + 1, seen)

    go(n - 1, 0, [])
    return total


# HELPERS for the base question
def step(ways):
    # next column from the current column only
    n = len(ways)
    return [(ways[r - 1] if r else 0) + ways[r]
            + (ways[r + 1] if r + 1 < n else 0)
            for r in range(n)]


def keep_row(ways, row):
    # a checkpoint forces the path through one row
```

```python
    return [x if r == row else 0
            for r, x in enumerate(ways)]


# FOLLOW-UP 1: Two checkpoints share a column.
# Answer: No path can visit both, because a path is in
# exactly one row per column, so the answer is 0 (the code
# returns it before the DP). Checkpoints at the start or
# end cell are fine: they only filter rows of the first or
# last column.
# NEEDED BY THE BASE QUESTION: it is also the answer to
# the follow-up below
def checkpoint_rows(checkpoints, ordered):
    # column -> required row; None if impossible
    rows = {}
```

```python
    for r, c in checkpoints:
        # two different checkpoints in one column: no path
        if rows.setdefault(c, r) != r:
            return None
    if len(rows) != len(checkpoints):
        return None
    cols = [c for _, c in checkpoints]
    # ordered: they must be listed left to right
    if ordered and cols != sorted(cols):
        return None
    return rows


# BASE QUESTION (optimal): Column DP with a rolling array
# and checkpoint filters
# Idea: ways[r] is the number of paths that are in row r
# at the current column. The next column is ways'[r] =
# ways[r - 1] + ways[r] + ways[r + 1]. At a checkpoint
# column keep only the checkpoint's row.
def count_paths(n, m, checkpoints=(), ordered=False):
    rows = checkpoint_rows(list(checkpoints), ordered)
    if rows is None:
        return 0
    # one rolling column is enough: O(n) space
    ways = [0] * n
    ways[n - 1] = 1
    if 0 in rows:
        ways = keep_row(ways, rows[0])
    for c in range(1, m):
        ways = step(ways)
        if c in rows:
            ways = keep_row(ways, rows[c])
    return ways[n - 1]


# HELPERS for the checks below
def count_paths_table(n, m):
    # full table: ways[c][r] paths reach row r, column c
    ways = [[0] * n for _ in range(m)]
    ways[0][n - 1] = 1
    for c in range(1, m):
        for r in range(n):
            # arrive from up-right, right or down-right
            for pr in (r - 1, r, r + 1):
                if 0 <= pr < n:
```

```python
                    ways[c][r] += ways[c - 1][pr]
    return ways[m - 1][n - 1]


if __name__ == "__main__":
    assert count_paths(3, 3) == 2
    assert count_paths_table(3, 3) == 2
    assert count_paths(1, 5) == 1
    assert count_paths(3, 1) == 1
    assert count_paths_table(4, 6) == count_paths(4, 6)
    # force the middle column through row 1
```

```
assert count_paths(3, 3, [(1, 1)]) == 1
# two checkpoints in one column: impossible
assert count_paths(3, 3, [(0, 1), (1, 1)]) == 0
# ordered checkpoints must go left to right
assert count_paths(3, 4, [(1, 1), (1, 2)], True) == 1
assert count_paths(3, 4, [(1, 2), (1, 1)], True) == 0
```

NOTES FROM CANDIDATES

Asked in an L4 phone interview.
Every move advances exactly one column, so the path visits each column exactly once — this is the key structural fact the follow-ups build on.
Clarify for follow-up 1 what happens if two checkpoints share a column (no path can visit both) and whether checkpoints are given as a list or marked in the grid.

PREPARATION

Write the column-by-column DP where dp[c][r] depends only on dp[c-1][r-1..r+1], then reduce it to a single rolling column.
For checkpoints, practice the "segment" approach: sort by column, and between consecutive checkpoints only allow paths that pass through them (zero out other rows in a checkpoint's column).
Test: n = 1 (only straight right), m = 1, a checkpoint at the start or end, and two checkpoints in the same column.

STUDY PATH: question 24 of 167 (Core 90)

Priority score 6.0 | Dynamic programming and memoization | asked 16 days ago | frequency Low

MEDIUM | 1Point3Acres | phone-screen, onsite-coding | 2 of 9

Longest Non-Decreasing Subarray with One Replacement

ASKED 2026-10-04 | 16 DAYS AGO | LAST 30 DAYS

STUDY: READ **NO LEETCODE FREQUENCY** **PRIORITY 209/228**

ABOUT THIS QUESTION

Topic: Core 90 - Dynamic programming and memoization

Optimal answer from: written and tested here for this exact question

Frequency: Low | Last asked: 2026-10-04 | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/longest-non-decreasing-subarray-one-replacement

QUESTION

Find the maximum length of a contiguous non-decreasing subarray if at most one element may be changed to any value. Some rounds start from the plain longest-increasing-subarray question and add the one-replacement follow-up; those versions ask for an increasing run, so clarify strict vs non-strict.

Input: an integer array nums.
You may change at most one element to any value.
Return the maximum length of a contiguous subarray that is non-decreasing (nums[i] <= nums[i + 1] for every adjacent pair inside it) after the change.

Variant: other rounds ask for a strictly increasing run and open with the no-replacement version (longest increasing subarray) before adding the one-replacement follow-up — clarify strict vs non-strict before coding.

Expect to discuss and compare the complexity of several approaches; a brute-force first pass is usually followed by a request to optimize.

EXAMPLES

```
nums = [1, 0, 3, 4, 5, 2, 3]
Output: 5
Change nums[1] from 0 to 1 -> [1, 1, 3, 4, 5, 2, 3]
[1, 1, 3, 4, 5] is a non-decreasing subarray of
    length 5.
```

APPROACH

Write the plain longest-run scan, then extend it with left-run / right-run arrays (or a two-state DP: "no change used" / "change used") so you can explain the jump from one to the other.
Test the bridge case explicitly: joining the run ending at i - 1 with the run starting at i + 1 is only valid when nums[i - 1] <= nums[i + 1]; in the strict variant there must be room for a value strictly between the neighbors, i.e. nums[i + 1] - nums[i - 1] >= 2 when the replacement must be an integer (plain < only if any real value is allowed). Otherwise the change only extends one side by one.
Repeat the exercise with the strict-increasing variant to make sure your comparisons flip correctly.

HOW TO THINK ABOUT IT

Precompute left[i], the length of the non-decreasing run ending at i, and right[i], the length of the run starting at i. If you change element i to any value, it can join the run on its left and the run on its right when a[i-1] <= a[i+1] (a value between them exists); otherwise it can join only the longer of the two. Take the best over all i.

Basis: Written for this question (the one-change version of LeetCode 674, Longest Continuous Increasing Subsequence).

BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def brute(a):
    best = longest_non_decreasing(a)
    for i in range(len(a)):
        for v in candidate_values(a):
            b = a[:i] + [v] + a[i + 1:]
            run = longest_non_decreasing(b)
            best = max(best, run)
    return best
```

TIME	SPACE
O(n^3)	O(n)

WHY IT IS SLOW

O(n^3). The run lengths on both sides of each position, computed once, answer every change in O(1).

OPTIMAL SOLUTION · Left and right run lengths

TIME	SPACE
O(n)	O(n)

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 The run must be strictly increasing.

Use < in the run lengths, and when the replaced value must fit between the neighbours check a[i+1] - a[i-1] >= 2 (an integer strictly between them exists) instead of a[i-1] <= a[i+1]. Clarify strict versus non-strict before coding.

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# MAIN QUESTION: Longest Non-Decreasing Subarray with One
# Replacement
#
# Find the maximum length of a contiguous non-decreasing
# subarray if at most one element may be changed to any
# value. Some rounds start from the plain longest-
# increasing-subarray question and add the one-replacement
# follow-up; those versions ask for an increasing run, so
# clarify strict vs non-strict. Input: an integer array
# nums. You may change at most one element to any value.
# Return the maximum length of a contiguous subarray that
# is non-decreasing (nums[i] <= nums[i + 1] for every
# adjacent pair inside it) after the change. Variant:
# other rounds ask for a strictly increasing run and open
# with the no-replacement version (longest increasing
# subarray) before adding the one-replacement follow-up —
# clarify strict vs non-strict before coding. Expect to
# discuss and compare the complexity of several
# approaches; a brute-force first pass is usually followed
# by a request to optimize.
#
# FOLLOW-UPS (answered below, in this order):
# 1. The run must be strictly increasing.
#
# INTERVIEW NOTES
# Precompute left[i], the length of the non-decreasing run
# ending at i, and right[i], the length of the run
# starting at i. If you change element i to any value, it
# can join the run on its left and the run on its right
# when a[i-1] <= a[i+1] (a value between them exists);
# otherwise it can join only the longer of the two. Take
# the best over all i.
#
# BRUTE FORCE (say it first): Try every change and re-scan
# For every position and every candidate value, rewrite
# that element and find the longest non-decreasing run.
# Correct and a good oracle; cubic. Time O(n^3). Space
# O(n).


# BRUTE FORCE (say it first): Try every change and re-scan
# | time O(n^3)
def one_change_brute(a):
    best = 0
    cands = set(a) | {0}
    for i in range(len(a)):
        for v in cands | {a[i]}:
            b = a[:i] + [v] + a[i + 1 :]
            run = 0
            for k in range(len(b)):
                run = (
                    run + 1
                    if k and b[k - 1] <= b[k]
                    else 1
                )
                best = max(best, run)
    return best


# HELPERS for the base question
def run_lengths(a, ok):
    n = len(a)
    # left[i]: run ending at i, right[i]: starting at i
    left, right = [1] * n, [1] * n
    for i in range(1, n):
        if ok(a[i - 1], a[i]):
            left[i] = left[i - 1] + 1
    for i in range(n - 2, -1, -1):
        if ok(a[i], a[i + 1]):
            right[i] = right[i + 1] + 1
    return left, right


# BASE QUESTION (optimal): Left and right run lengths
# Idea: left[i] and right[i] are the non-decreasing run
# lengths ending and starting at i. Changing a[i] gives
# left[i-1] + 1 + right[i+1] if a[i-1] <= a[i+1],
# otherwise max(left[i-1], right[i+1]) + 1.
# FOLLOW-UP 1: The run must be strictly increasing.
# Answer: Use < in the run lengths, and when the replaced
# value must fit between the neighbours check a[i+1] -
# a[i-1] >= 2 (an integer strictly between them exists)
# instead of a[i-1] <= a[i+1]. Clarify strict versus non-
# strict before coding.
def longest_with_one_change(a, strict=False):
    n = len(a)
    # nothing to fix in arrays of length 0, 1 or 2
    if n <= 2:
        return n
    # FOLLOW-UP 1: strict=True means strictly increasing
    # non-decreasing uses <=, strictly increasing uses <
    def ok(x, y):
        return x < y if strict else x <= y

    # can a value fit between the two neighbours?
    gap = 2 if strict else 0
    left, right = run_lengths(a, ok)
    best = 1
    # try changing each element a[i]
    for i in range(n):
        L = left[i - 1] if i else 0
        R = right[i + 1] if i + 1 < n else 0
        # at an end, or when a value fits between the
        # neighbours, the change joins both runs
        if not L or not R or a[i + 1] - a[i - 1] >= gap:
            best = max(best, L + 1 + R)
        # otherwise it joins only the longer run
        else:
            best = max(best, max(L, R) + 1)
    return best


# WARM-UP: no replacement allowed
def longest_without_change(a):
    # the plain longest increasing subarray warm-up
    best = run = 0
    for i, x in enumerate(a):
        run = run + 1 if i and a[i - 1] < x else 1
        best = max(best, run)
    return best


if __name__ == "__main__":
    # change 0 to 1: [1, 1, 3, 4, 5] has length 5
    nums = [1, 0, 3, 4, 5, 2, 3]
    assert longest_with_one_change(nums) == 5
    # tiny arrays
    assert longest_with_one_change([]) == 0
```

```
assert longest_with_one_change([7]) == 1
# strict version needs room between neighbours
strict = longest_with_one_change
assert strict([1, 2, 2, 3, 4], True) == 4
assert strict([1, 1, 1, 1], True) == 2
# the warm-up without any change
assert longest_without_change([1, 3, 5, 4, 7]) == 3
```

NOTES FROM CANDIDATES

Asked as an onsite coding round in an early-career L3 loop, and as the first technical round of new-grad VOs.
Candidates who wrote a slow version first were asked to optimize, leaving little time for anything else — the problem is short, so interviewers expect the linear approach to arrive quickly.
Edge cases worth confirming: arrays of length 0–2, where the replaced element is the first or last of the run, and whether the replaced value must fit between both neighbors or only one.

PREPARATION

Write the plain longest-run scan, then extend it with left-run / right-run arrays (or a two-state DP: "no change used" / "change used") so you can explain the jump from one to the other.
Test the bridge case explicitly: joining the run ending at i - 1 with the run starting at i + 1 is only valid when nums[i - 1] <= nums[i + 1]; in the strict variant there must be room for a value strictly between the neighbors, i.e. nums[i + 1] - nums[i - 1] >= 2 when the replacement must be an integer (plain < only if any real value is allowed). Otherwise the change only extends one side by one.
Repeat the exercise with the strict-increasing variant to make sure your comparisons flip correctly.

STUDY PATH: question 25 of 167 (Core 90)

Priority score 5.5 | Dynamic programming and memoization | asked 4 days ago | frequency Single report

MEDIUM | 1Point3Acres | tech-screen | 3 of 9

Bounded Knapsack Variant

ASKED 2026-10-16 | 4 DAYS AGO | LAST 30 DAYS

STUDY: TALK **NO LEETCODE FREQUENCY** **PRIORITY 180/228**

ABOUT THIS QUESTION

Topic: Core 90 - Dynamic programming and memoization

Optimal answer from: written and tested here for this exact question

LeetCode match: LC 1710 Maximum Units on a Truck

Frequency: Single report | Last asked: 2026-10-16 | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/unit-weight-bounded-knapsack

QUESTION

Each item type has a value and an available count; every item weighs 1. Given capacity W, maximize total value. DP or DFS + memo works at O(n·W), greedy by value after sorting is O(n log n), and the best answer uses quickselect for O(n).

Input: for each item type, its value and the number of copies available. Every item has weight 1.
Input: a maximum total weight W.
Output: the maximum total value achievable without exceeding W.

APPROACH

Write the sort-based greedy and argue its optimality in two sentences (unit weights mean every slot is interchangeable). Implement quickselect that partitions item types by value and tracks the total count on each side to locate where W runs out. Compare the O(n × W), O(n log n), and O(n) versions out loud with their complexities.

INTERVIEW-DAY PLAN

CLARIFY FIRST

Ask whether every item really weighs 1, whether values can be zero or negative (assume non-negative), whether W can exceed the total number of items, and how large the counts and W can get.

BRUTE FORCE TO SAY OUT LOUD

A knapsack table over the capacity, one pass per copy. O(total copies * W), correct for any weights.

OPTIMAL IDEA

With unit weights every slot is interchangeable, so sort the item types by value, highest first, and take min(copies, slots left) of each until the bag is full. O(n log n). To reach O(n), run quickselect on the value: split the types into higher, equal and lower, recurse into the side that contains the W-th slot, and keep a running total of what was taken.

INTERVIEW-DAY PLAN

DRY-RUN THESE CASES

W = 0, W larger than all copies together, one type, equal values, a type with a huge count, all values equal.

LIKELY FOLLOW-UPS

Different weights (bounded knapsack with binary splitting), a huge W (greedy still works), or negative values (never take them).

PITFALLS

Reaching for DP because the word knapsack appears. Forgetting that a type may only be partly taken. Letting the remaining capacity go negative.

HOW TO THINK ABOUT IT

The word knapsack points at DP, but the statement says every item weighs 1. With equal weights every slot in the bag is interchangeable, so an item that is worth more is never worse than one that is worth less: swapping a cheaper item for a dearer one always helps. Take item types by value, highest first, until W slots are used. You do not even need the full sort: quickselect finds where the W-th slot falls in O(n).

Basis: LeetCode 1710, Maximum Units on a Truck: every item weighs 1, so the knapsack collapses to 'take the most valuable items first'.

BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def knapsack(items, W):
    best = [0] * (W + 1)
    for value, count in items:
        for _ in range(count):
            for w in range(W, 0, -1):
                take = best[w - 1] + value
                best[w] = max(best[w], take)
    return best[W]
```

TIME	SPACE
O(total copies * W)	O(W)

WHY IT IS SLOW

Every copy is its own pass over the table, O(total copies * W). With unit weights the most valuable items are always the right ones, so sorting by value and taking greedily is O(n log n).

OPTIMAL SOLUTION · Sort by value and take greedily

TIME	SPACE
O(n log n)	O(1) extra

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 Why is greedy correct here but not for a general knapsack?

With unit weights each chosen item uses exactly one slot, so the only thing that matters is value per slot, which is just the value. If an optimal bag held a cheaper item while a dearer one was available, swapping them keeps the weight and raises the value, so the optimal bag can always be rearranged into the greedy one. With different weights a cheap item can leave room for several others, so value per weight and the leftover space interact and greedy fails.

FOLLOW-UP 2 What if the weights are not all 1?

Then it is a real bounded knapsack and greedy no longer works; use DP over the capacity. Splitting each count into 1, 2, 4, ... copies keeps the number of 0/1 items at about log(count) per type, so the cost is O(n * log(max count) * W) instead of O(total copies * W).

FOLLOW-UP 3 W is up to 10^9. Which version still works?

Only the greedy ones: sorting is O(n log n) and quickselect O(n), and neither depends on W, while any table over the capacity needs O(W) memory. The loop simply stops when the remaining capacity reaches zero.

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# MAIN QUESTION: Bounded Knapsack Variant
#
# Each item type has a value and an available count; every
# item weighs 1. Given capacity W, maximize total value.
# DP or DFS + memo works at O(n·W), greedy by value after
# sorting is O(n log n), and the best answer uses
# quickselect for O(n). Input: for each item type, its
# value and the number of copies available. Every item has
# weight 1. Input: a maximum total weight W. Output: the
# maximum total value achievable without exceeding W.
#
# FOLLOW-UPS (answered below, in this order):
# 1. Why is greedy correct here but not for a general
# knapsack?
# 2. What if the weights are not all 1?
# 3. W is up to 10^9. Which version still works?
#
# INTERVIEW NOTES
# The word knapsack points at DP, but the statement says
# every item weighs 1. With equal weights every slot in
# the bag is interchangeable, so an item that is worth
# more is never worse than one that is worth less:
# swapping a cheaper item for a dearer one always helps.
# Take item types by value, highest first, until W slots
# are used. You do not even need the full sort:
# quickselect finds where the W-th slot falls in O(n).
#
# BRUTE FORCE (say it first): DP over the capacity
# Treat every copy as its own 0/1 item and run the classic
# 1D knapsack table. It is correct for any weights, so it
# is the oracle, but it is far too slow when the counts or
# W are large. Time O(total copies * W). Space O(W).


import random


# BRUTE FORCE (say it first, it also checks the answers)
def knapsack_dp(items, W):
    # the oracle: every copy is its own 0/1 item
    best = [0] * (W + 1)
    for value, count in items:
        for _ in range(count):
            for w in range(W, 0, -1):
                take = best[w - 1] + value
                best[w] = max(best[w], take)
    return best[W]


# BASE QUESTION (optimal): Sort by value and take greedily
# Idea: Sort the item types by value, highest first. For
# each type take min(copies, slots left) and add value
# times that. Stop when the bag is full.
def max_value_sorted(items, W):
    """Unit weights: every slot is worth the same, so the
    best value always goes in first."""
    total = 0
    for value, count in sorted(items, reverse=True):
        # take as many copies as still fit in the bag
        take = min(count, W)
        total += value * take
        W -= take
        if W == 0:
            break
    return total


# SAME answer in O(n): quickselect
def max_value_select(items, W):
    """Same answer in O(n) expected: quickselect finds
    where the W-th slot falls without a full sort."""
    total, pool = 0, list(items)
    while pool and W > 0:
        pivot = random.choice(pool)[0]
        high = [it for it in pool if it[0] > pivot]
        mid = [it for it in pool if it[0] == pivot]
        low = [it for it in pool if it[0] < pivot]
        high_count = sum(c for _, c in high)
        if high_count >= W:
            # the bag fills up inside the dearer types
            pool = high
            continue
        # every dearer copy fits, then fill from the pivot
        total += sum(v * c for v, c in high)
        W -= high_count
        take = min(W, sum(c for _, c in mid))
        total += pivot * take
        W -= take
        pool = low
    return total


# FOLLOW-UP 2: What if the weights are not all 1?
# Answer: Then it is a real bounded knapsack and greedy no
# longer works; use DP over the capacity. Splitting each
# count into 1, 2, 4, ...
def knapsack_bounded(items, W):
    """Weights differ: greedy fails, so use DP. Splitting
    each count into 1, 2, 4, ... copies keeps it small."""
    best = [0] * (W + 1)
    for value, weight, count in items:
        chunk = 1
```

```
        while count > 0:
            take = min(chunk, count)
            count -= take
            cost, gain = take * weight, take * value
            for w in range(W, cost - 1, -1):
                best[w] = max(best[w],
                              best[w - cost] + gain)
            chunk *= 2
    return best[W]


if __name__ == "__main__":
    demo = [(3, 2), (5, 1), (4, 4)]
    assert max_value_sorted(demo, 4) == 17
    assert max_value_select(demo, 4) == 17
    assert max_value_sorted([], 5) == 0
    assert max_value_sorted([(7, 3)], 0) == 0
    assert max_value_sorted([(7, 3)], 10) == 21
    mixed = [(10, 5, 1), (6, 3, 2)]
    assert knapsack_bounded(mixed, 6) == 12
    # greedy is only valid when every weight is 1
    unit = [(3, 1, 2), (5, 1, 1), (4, 1, 4)]
    assert knapsack_bounded(unit, 4) == 17
    for _ in range(300):
        n = random.randint(0, 6)
        rnd = random.randint
        items = [(rnd(1, 9), rnd(1, 4)) for _ in range(n)]
        W = random.randint(0, 12)
        want = knapsack_dp(items, W)
        assert max_value_sorted(items, W) == want
        assert max_value_select(items, W) == want
```

NOTES FROM CANDIDATES

Pitfalls: Recognizing "knapsack" and reaching for DP is the trap; with unit weights the greedy argument is simpler and faster.

PREPARATION

Write the sort-based greedy and argue its optimality in two sentences (unit weights mean every slot is interchangeable). Implement quickselect that partitions item types by value and tracks the total count on each side to locate where W runs out. Compare the O(n × W), O(n log n), and O(n) versions out loud with their complexities.

STUDY PATH: question 26 of 167 (Core 90)

Priority score 4.96 | Dynamic programming and memoization | asked 68 days ago | frequency Single report

MEDIUM | 1Point3Acres | onsite-coding | 4 of 9

Target Expressions from Ordered Digits

ASKED 2026-08-13 | 68 DAYS AGO | LAST 90 DAYS

STUDY: TALK | **ALL TIME 36.5** | **30 DAYS 0** | **PRIORITY 103/228**

ABOUT THIS QUESTION

Topic: Core 90 - Dynamic programming and memoization

Optimal answer from: written and tested here for this exact question

LeetCode match: LC 282 Expression Add Operators

Frequency: Single report | Last asked: 2026-08-13 | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/target-expressions-ordered-digits

QUESTION

EXAMPLES

```
For digits [2, 3, 4] and target 14, valid outputs
    include:
2 * (3 + 4)
2 + 3 * 4
```

APPROACH

Enumerate operator placements and valid parenthesizations for three or four ordered digits.
Explore memoized dynamic-programming state definitions for the N-digit follow-up and be ready to justify one before coding.
Explain how duplicate expressions or duplicate values would be handled without discarding distinct valid output forms.

INTERVIEW-DAY PLAN

CLARIFY FIRST

Ask whether digits may be concatenated into multi-digit numbers, which operators are allowed, whether parentheses are free, whether the digit order is fixed, and whether to return all expressions or a boolean.

BRUTE FORCE TO SAY OUT LOUD

Enumerate every operator placement and every parenthesization for the three or four digits and evaluate each. That is fine for a handful of digits and is what the first stage asks for.

OPTIMAL IDEA

Interval DP: for each interval of digits compute a map from value to the expressions that achieve it by splitting at every point and combining left and right with plus, minus and times. Memoize by interval. If only reachability is needed store just the set of values per interval, which is much smaller than the strings.

DRY-RUN THESE CASES

[2, 3, 4] with target 14 gives 2*(3+4) and 2+3*4. A target that is unreachable. A single digit. Zeros.

INTERVIEW-DAY PLAN

LIKELY FOLLOW-UPS

N digits (the DP above), division with fractions, or concatenation.

PITFALLS

Missing parentheses in the output string. Evaluating with floats.

HOW TO THINK ABOUT IT

Any expression over a run of digits is two smaller expressions joined by an operator at some split point, so the set of values for an interval is built from the sets of its two halves. Compute values bottom-up per interval and only build text for the target. Be explicit about parentheses: they come from the split structure, so every split is a candidate.

Basis: Related to LeetCode 282, Expression Add Operators, but with parentheses allowed and digits not concatenated.

BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def values(digits):
    if len(digits) == 1:
        return {digits[0]}
    result = set()
    for cut in range(1, len(digits)):
        for a in values(digits[:cut]):
            for b in values(digits[cut:]):
                result |= {a + b, a - b, a * b}
    return result
reachable = target in values(all_digits)
```

TIME	SPACE
exponential (Catalan(n) *...	**O(n)**
exponential (Catalan(n) * 3^(n-1) expressions)	recursion plus the value set

WHY IT IS SLOW

The same sub-interval of digits is recomputed from scratch for every split above it, giving Catalan-number growth. Remembering the values per interval (memoisation) makes it polynomial in the number of distinct values.

OPTIMAL SOLUTION · Interval DP: all expressions for digits[l:r]

TIME	SPACE
exponential in the number...	**O(distinct values per interval)**
exponential in the number of results; polynomial in the number of distinct values	

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 How do you handle N digits where listing every expression is too much?

Agree on the contract: if you only need to know whether a target is reachable, store just the set of values per interval (a boolean DP over (l, r, value)), which is far smaller than the set of strings. If you need one witness, store one expression string per value. Values are bounded by the digits, so the table stays practical for moderate N.

FOLLOW-UP 2 What if concatenating digits (23) or division is allowed?

Concatenation adds base cases: for every l and r the number int(digits[l:r]) is also an atom of the interval. Division adds a fourth operator with a zero check and, if exactness matters, Fraction values instead of floats.

FOLLOW-UP 3 Only a yes/no answer is needed for many digits.

Store just the set of reachable values per interval (as target_reachable does) instead of the expression strings; the strings blow up long before the value sets do. Build the expression only for the winning target by remembering one split per value.

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# MAIN QUESTION: Target Expressions from Ordered Digits
#
# Examples: For digits [2, 3, 4] and target 14, valid
# outputs include: 2 * (3 + 4) 2 + 3 * 4
#
# FOLLOW-UPS (answered below, in this order):
# 1. How do you handle N digits where listing every
# expression is too much?
# 2. What if concatenating digits (23) or division is
# allowed?
# 3. Only a yes/no answer is needed for many digits.
#
# INTERVIEW NOTES
# Any expression over a run of digits is two smaller
# expressions joined by an operator at some split point,
# so the set of values for an interval is built from the
# sets of its two halves. Compute values bottom-up per
# interval and only build text for the target. Be explicit
# about parentheses: they come from the split structure,
# so every split is a candidate.
#
# BRUTE FORCE (say it first): Recursively try every split
# and operator, without remembering sub-results
# Split the digits at every point, combine all value pairs
# of the two parts with +, - and *, and see whether the
# target appears. Correct, but the same sub-interval is
# recomputed many times; memoising by interval is the fix.
# Time exponential (Catalan(n) * 3^(n-1) expressions).
# Space O(n) recursion plus the value set.
```

```
from collections import defaultdict


# BRUTE FORCE (say it first): Recursively try every split
# and operator, without remembering sub-results | time
# exponential (Catalan(n) * 3^(n-1) expressions)
def _all_values(digits):
    if len(digits) == 1:
        return {digits[0]}
    out = set()
    for i in range(1, len(digits)):
        for a in _all_values(digits[:i]):
            for b in _all_values(digits[i:]):
                out |= {a + b, a - b, a * b}
    return out


def target_reachable_brute(digits, target):
    return target in _all_values(digits)


# HELPERS for the base question
def build(digits):
    # table[(l, r)] = {value: {(text, precedence)}}
    table = {}

    def solve(l, r):
        if (l, r) in table:
            return table[(l, r)]
        res = defaultdict(set)
        if r - l == 1:
            # a single digit is an atom (precedence 3)
            res[digits[l]].add((str(digits[l]), 3))
        # try every split point
        for m in range(l + 1, r):
            for lv, ls in solve(l, m).items():
                for rv, rs in solve(m, r).items():
                    combine(res, lv, ls, rv, rs)
        table[(l, r)] = res
        return res

    return solve(0, len(digits))


def combine(res, lv, ls, rv, rs):
    ops = (("+", 1, lv + rv), ("-", 1, lv - rv),
           ("*", 2, lv * rv))
    for lt, lp in ls:
        for rt, rp in rs:
            for op, prec, val in ops:
                # brackets only where precedence needs
                left = f"({lt})" if lp < prec else lt
                need = rp < prec
                if rp == prec and op == "-":
                    need = True
                right = f"({rt})" if need else rt
                res[val].add((left + op + right, prec))
```

```
# BASE QUESTION (optimal): Interval DP: all expressions
# for digits[l:r]
# Idea: Every expression over digits[l:r] is two sub-
# expressions over a split point joined by +, - or *.
# Memoize, for each interval, a map from numeric value to
# the set of expression strings that produce it.
def expressions_for_target(digits, target):
    found = build(digits).get(target, ())
    return sorted({text for text, _ in found})


# FOLLOW-UP 3: Only a yes/no answer is needed for many
# digits.
```

```python
# Answer: Store just the set of reachable values per
# interval (as target_reachable does) instead of the
# expression strings; the strings blow up long before the
# value sets do.
def target_reachable(digits, target):
    # yes/no only: keep just the set of values per range
    values = {}

    def solve(l, r):
        if (l, r) in values:
            return values[(l, r)]
        out = {digits[l]} if r - l == 1 else set()
        for m in range(l + 1, r):
            for a in solve(l, m):
                for b in solve(m, r):
                    out |= {a + b, a - b, a * b}
        values[(l, r)] = out
        return out

    return target in solve(0, len(digits))


if __name__ == "__main__":
    ans = expressions_for_target([2, 3, 4], 14)
    assert "2*(3+4)" in ans and "2+3*4" in ans
    assert target_reachable([2, 3, 4], 14)
    assert not target_reachable([2, 3, 4], 100)
    # subtraction needs brackets on the right
    sub = expressions_for_target([5, 3, 1], 3)
    assert "5-(3-1)" in sub
    assert target_reachable([1] * 8, 8)
```

NOTES FROM CANDIDATES

The base case is small enough for direct enumeration, but a complicated ad hoc implementation becomes difficult to extend. Keep expression construction separate from numeric evaluation while negotiating the N-digit contract.
If the agreed contract uses every digit exactly once in order, memoize each interval [l, r) as a map from attainable numeric values to the expression strings producing them. For every split point, combine left and right entries with + and *, wrapping composite children as needed; the full-interval target bucket is the answer. Because every distinct expression must be retained, both time and storage are output-sensitive and can grow exponentially even though interval states are memoized.

PREPARATION

Enumerate operator placements and valid parenthesizations for three or four ordered digits.
Explore memoized dynamic-programming state definitions for the N-digit follow-up and be ready to justify one before coding.
Explain how duplicate expressions or duplicate values would be handled without discarding distinct valid output forms.

STUDY PATH: question 27 of 167 (Core 90)

Priority score 4.5 | Dynamic programming and memoization | asked 41 days ago | frequency Single report

MEDIUM | 1Point3Acres | phone-screen | 5 of 9

Minimum-Cost Tree Disconnection

ASKED 2026-09-09 | 41 DAYS AGO | LAST 90 DAYS

STUDY: TALK | **NO LEETCODE FREQUENCY** | **PRIORITY 197/228**

ABOUT THIS QUESTION

Topic: Core 90 - Dynamic programming and memoization

Optimal answer from: written and tested here for this exact question

Frequency: Single report | Last asked: 2026-09-09 | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/binary-tree-root-leaf-minimum-cut

QUESTION

EXAMPLES

Negative-weight follow-up

```
A --(-1)--> B --(-1)--> C
Minimum total cut cost: -2
```

Cutting both edges costs -2, even though cutting just the first edge already separates the root from the original leaf.

APPROACH

Implement the nonnegative postorder recurrence, then trace a chain and a branching tree while accounting for recursion depth. Implement the two-state negative-weight version and compare it with exhaustive edge-subset enumeration on small trees, including the given negative chain.

INTERVIEW-DAY PLAN

CLARIFY FIRST

Ask what must be disconnected (root from every leaf), whether the tree is rooted and directed, whether weights can be negative, and what to return for an empty or single-node tree.

BRUTE FORCE TO SAY OUT LOUD

Try every subset of edges and keep those that separate the root from all leaves. Exponential.

OPTIMAL IDEA

Define best(v) as the cheapest set of cuts in v's subtree that disconnects v from every leaf below it. For each child c with edge weight w, either cut the edge, paying w, or keep it and require best(c) (not allowed if c is a leaf). Sum over children. With non-negative weights that is the answer. With negative weights, once you cut an edge the whole subtree below becomes free, and you may still cut any negative edge there, so the cut option costs w plus the sum of min(0, edge) over that subtree.

DRY-RUN THESE CASES

A single edge, the negative chain A to B to C (answer -2), a leaf at different depths, a star.

LIKELY FOLLOW-UPS

Negative weights, many roots, or recursion depth for a chain tree.

PITFALLS

Treating every negative edge as a mandatory cut. Recursing deeply on a chain.

HOW TO THINK ABOUT IT

On a tree, every decision about a subtree depends only on the edge above it, so define a function of the node that returns the best cost inside its subtree. For each child you either cut the edge above it (pay its weight) or keep it and make the child itself disconnected from the leaves below. With negative weights, a cut edge frees the whole subtree, so you may cut every negative edge in it for free gain.

Basis: Written. Assumed setup (confirm it): a rooted tree with edge weights (cuts cost the weight, weights may be negative); cut edges so the root is disconnected from every leaf; a leaf is a non-root node with no children; with no leaves the cost is 0.

BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def brute(root, edges):
    best = INF
    for k in range(len(edges) + 1):
        for cut in combinations(edges, k):
            tree = remove_edges(edges, cut)
            if not reaches_leaf(tree, root):
                cost = sum(w for _, _, w in cut)
                best = min(best, cost)
    return best
```

TIME	SPACE
O(2^E * n)	O(n)

WHY IT IS SLOW

2^E subsets, each checked with a search: O(2^E * n). A tree DP (cut the edge above a child, or require the child to be disconnected from its leaves) is O(n).

OPTIMAL SOLUTION · Tree DP: best(v) = cheapest cuts inside v's subtree that disconnect v from its leaves

TIME	SPACE
O(n)	O(h)

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 Weights are non-negative.

Then free(c) is always 0 and the recurrence collapses to best(v) = sum over children of (w if the child is a leaf, else min(w, best(child))).

FOLLOW-UP 2 The tree is a long chain and recursion overflows.

Compute nodes in reverse BFS order (children before parents) and store best and free in arrays, no recursion needed. State the space as O(n).

FOLLOW-UP 3 Return which edges to cut.

Remember at each child which option was cheaper, then walk down from the root collecting the cut edges. Cut edges also include every negative edge below any cut edge.

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# MAIN QUESTION: Minimum-Cost Tree Disconnection
#
# Examples: Negative-weight follow-up: A --(-1)--> B
# --(-1)--> C Minimum total cut cost: -2 Cutting both
# edges costs -2, even though cutting just the first edge
# already separates the root from the original leaf.
#
# FOLLOW-UPS (answered below, in this order):
# 1. Weights are non-negative.
# 2. The tree is a long chain and recursion overflows.
# 3. Return which edges to cut.
#
# INTERVIEW NOTES
# On a tree, every decision about a subtree depends only
# on the edge above it, so define a function of the node
# that returns the best cost inside its subtree. For each
# child you either cut the edge above it (pay its weight)
# or keep it and make the child itself disconnected from
# the leaves below. With negative weights, a cut edge
# frees the whole subtree, so you may cut every negative
# edge in it for free gain.
#
# BRUTE FORCE (say it first): Try every subset of edges
# Enumerate each subset of edges to cut, check that the
# root can reach no leaf, and keep the cheapest.
# Exponential, but exact, so it checks the DP. Time O(2^E
# * n). Space O(n).


from collections import defaultdict


# BRUTE FORCE (say it first): Try every subset of edges |
# time O(2^E * n)
def disconnect_brute(n, edges):
    children = defaultdict(list)
    for u, v, w in edges:
        children[u].append(v)
    leaves = [
        v
        for v in range(n)
        if v != 0 and not children[v]
    ]
    best = None
    for mask in range(1 << len(edges)):
        kept = [
            (u, v)
            for i, (u, v, w) in enumerate(edges)
            if not mask >> i & 1
        ]
        adj = defaultdict(list)
        for u, v in kept:
            adj[u].append(v)
        seen, stack = {0}, [0]
        while stack:
            x = stack.pop()
            for y in adj[x]:
                if y not in seen:
                    seen.add(y)
                    stack.append(y)
        if not any(leaf in seen for leaf in leaves):
            cost = sum(
                w
                for i, (u, v, w) in enumerate(edges)
                if mask >> i & 1
            )
            best = (
                cost
                if best is None
                else min(best, cost)
            )
    return best if best is not None else 0


# HELPERS for the base question
INF = float("inf")


def tree_costs(n, edges):
    # edges are (parent, child, cost); node 0 is the root
    kids = defaultdict(list)
    for u, v, w in edges:
        kids[u].append((v, w))
    # FOLLOW-UP 2: a loop, not recursion: safe on a long
    # chain
    # breadth-first order: parents come before children
    order = [0]
    for u in order:
        for v, _ in kids[u]:
            order.append(v)
    # best[v]: cheapest way to cut v off from its leaves
    # free[v]: sum of every negative cost below v
    best, free = [0] * n, [0] * n
    for v in reversed(order):
        for c, w in kids[v]:
            # 1) cut this edge and the negatives below
            cut = w + free[c]
            # 2) keep it and cut deeper (a leaf cannot)
            keep = best[c] if kids[c] else INF
            best[v] += min(cut, keep)
            free[v] += min(0, w) + free[c]
    return kids, best, free


# BASE QUESTION (optimal): Tree DP: best(v) = cheapest
# cuts inside v's subtree that disconnect v from its
# leaves
# Idea: For a child c with edge weight w there are two
# options: cut the edge, paying w plus whatever extra
# negative-weight edges you can still cut below c (free(c)
# = sum of min(0, weight) over its subtree), or keep the
# edge and require best(c), which is impossible when c is
# a leaf.
```

```python
# FOLLOW-UP 1: Weights are non-negative.
# Answer: Then free(c) is always 0 and the recurrence
# collapses to best(v) = sum over children of (w if the
# child is a leaf, else min(w, best(child))).
# FOLLOW-UP 2: The tree is a long chain and recursion
# overflows.
# Answer: Compute nodes in reverse BFS order (children
# before parents) and store best and free in arrays, no
# recursion needed. State the space as O(n).
def disconnect_cost(n, edges):
    # works for negative costs and needs no recursion
    return tree_costs(n, edges)[1][0]


# FOLLOW-UP 3: Return which edges to cut.
# Answer: Remember at each child which option was cheaper,
# then walk down from the root collecting the cut edges.
# Cut edges also include every negative edge below any cut
# edge.
def edges_to_cut(n, edges):
    kids, best, free = tree_costs(n, edges)
    cuts, stack = [], [0]
```

```python
    while stack:
        v = stack.pop()
        for c, w in kids[v]:
            keep = best[c] if kids[c] else INF
            if w + free[c] <= keep:
                # cut here, plus negative edges below c
                cuts.append((v, c))
                below = [c]
                while below:
                    x = below.pop()
                    for y, wy in kids[x]:
                        if wy < 0:
                            cuts.append((x, y))
                        below.append(y)
            else:
                stack.append(c)
    return cuts


if __name__ == "__main__":
    # the question's chain A -(-1)-> B -(-1)-> C
    neg = [(0, 1, -1), (1, 2, -1)]
    assert disconnect_cost(3, neg) == -2
    cuts = sorted(edges_to_cut(3, neg))
    assert cuts == [(0, 1), (1, 2)]
    # non-negative costs: cut the cheaper edge
    chain = [(0, 1, 5), (1, 2, 3)]
    assert disconnect_cost(3, chain) == 3
    assert edges_to_cut(3, chain) == [(1, 2)]
    # a single node needs no cut
    assert disconnect_cost(1, []) == 0
    # a long chain does not overflow the stack
    big = [(i, i + 1, 1) for i in range(5000)]
    assert disconnect_cost(5001, big) == 1
```

NOTES FROM CANDIDATES

Clarify the required result for an empty tree and a single-node tree. No final return convention is specified for these cases.
Include recursion-stack memory in space analysis. The tree can degenerate into a chain; balanced-tree depth is not a worst-case bound.
Coding took place directly in Google Docs with no supplied framework; the candidate also had to define the tree node structure.
The interviewer asked the candidate to think aloud while developing the approach.
For nonnegative weights, use postorder DP. Let F(v) be the minimum cost to disconnect node v from every original leaf below it. An original leaf has F(v) = infinity; an internal node has F(v) = sum(min(weight(v,c), F(c))) over its children. Each branch must be blocked independently. Use a safe infinity representation rather than adding to a fixed-width maximum integer.
For negative weights, cutting an ancestor edge does not remove the opportunity to cut negative edges below it. Track A(v), the minimum optional cut cost when the subtree is already disconnected: A(v) = sum(min(0, weight(v,c)) + A(c)), with A(leaf) = 0. Then use F(v) = sum(min(weight(v,c) + A(c), F(c))), retaining F(leaf) = infinity. This includes beneficial redundant negative cuts

without double counting edges.
Both recurrences process each edge once: O(n) time and O(h) recursive auxiliary space, where h can equal n. A single original root-leaf cannot be disconnected from itself by cutting edges; agree on its result convention before coding.

PREPARATION

Implement the nonnegative postorder recurrence, then trace a chain and a branching tree while accounting for recursion depth. Implement the two-state negative-weight version and compare it with exhaustive edge-subset enumeration on small trees, including the given negative chain.

STUDY PATH: question 28 of 167 (Core 90)

Priority score 4.24 | Dynamic programming and memoization | asked 98 days ago | frequency Single report

MEDIUM | 1Point3Acres | phone-screen | 6 of 9

Count Word Decompositions into Chemical Element Symbols

ASKED 2026-07-14 | 98 DAYS AGO | LAST 6 MONTHS

STUDY: TYPE | **ALL TIME 43.5** | **30 DAYS 0** | **PRIORITY 81/228**

ABOUT THIS QUESTION

Topic: Core 90 - Dynamic programming and memoization

Optimal answer from: written and tested here for this exact question

Related in your app (a neighbouring problem, not this exact question): LeetCode #139 Word Break

LeetCode match: LC 139 Word Break (related)

Frequency: Single report | Last asked: 2026-07-14 | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/periodic-table-word-decomposition

QUESTION

EXAMPLES

Physics has four stated decompositions

P - H - Y - Si - C - S P - H - Y - Si - Cs P - H - Y - S - I - C - S P - H - Y - S - I - Cs The returned count is 4.

APPROACH

Implement a suffix-count or prefix-count formulation and state what each state represents.

Test a word with no decomposition, one decomposition, multiple decompositions, and overlapping one- and two-letter symbols.

INTERVIEW-DAY PLAN

CLARIFY FIRST

Ask whether matching is case-insensitive, whether to count decompositions or just test for one, and for the dictionary (the periodic table symbols of one and two letters).

BRUTE FORCE TO SAY OUT LOUD

Recursive search that tries both a one-letter and a two-letter symbol at each position. Exponential in the worst case.

OPTIMAL IDEA

Suffix-count DP: ways[i] is the number of decompositions of word[i:], with ways[n] = 1. ways[i] adds ways[i + 1] if word[i] is a symbol and ways[i + 2] if word[i:i + 2] is a symbol. O(n). Physics gives 4.

DRY-RUN THESE CASES

Physics (4), a word with no decomposition (hello), one with exactly one, the empty string, overlapping one- and two-letter symbols such as Si versus S + I.

LIKELY FOLLOW-UPS

List the decompositions, handle a larger dictionary with a trie, or count modulo a prime.

INTERVIEW-DAY PLAN

PITFALLS

Forgetting to lowercase before the lookup. Starting the DP with ways[n] = 0.

HOW TO THINK ABOUT IT

Counting the ways to cut a string into dictionary pieces is Word Break with a count instead of a yes/no. Define ways[i] as the number of decompositions of the suffix starting at i; it depends only on ways[i + 1] and ways[i + 2], because symbols have one or two letters. Compute from the end so every value you need already exists.

Basis: A counting version of LeetCode 139, Word Break, with the periodic-table symbols as the dictionary.

BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
def ways(i):
    if i == len(word):
        return 1
    total = 0
    if word[i] in SYMBOLS:
        total += ways(i + 1)
    if word[i:i + 2] in SYMBOLS:
        total += ways(i + 2)
    return total
```

TIME	SPACE
exponential (Fibonacci gr...	**O(n)**
exponential (Fibonacci growth)	

WHY IT IS SLOW

The same suffix is reached by many different splits, so the calls grow like the Fibonacci numbers. Computing ways from the end of the word once per position is O(n).

OPTIMAL SOLUTION · Suffix-count DP

TIME	SPACE
O(n)	**O(n)**

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 How would you list the decompositions, not just count them?

Keep the same suffix recursion but return the list of splits instead of a number, memoized by start index. The count version is O(n); the list version is output-sensitive because there can be exponentially many splits, so agree on whether the interviewer wants a count, one witness, or all of them.

FOLLOW-UP 2 What if the dictionary is large and symbols can be long?

Put the dictionary in a trie (or a set plus the maximum word length L) and only look at substrings up to length L from each index. That makes the DP O(n * L) instead of O(n^2), which is the standard Word Break optimization.

FOLLOW-UP 3 List the decompositions instead of counting them.

Same recursion, but carry the parts chosen so far and record them at the end. The number of results can be exponential, so ask whether a count or a list is wanted.

FOLLOW-UP 4 A larger dictionary with long entries.

Store the dictionary in a set plus the maximum word length L (only look at substrings up to L), or in a trie, so each position costs O(L) lookups instead of O(n).

FOLLOW-UP 5 Count modulo a prime, or only test whether any decomposition exists.

Replace addition by addition modulo the prime, or by logical or for a yes/no answer. The DP structure is the same.

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# MAIN QUESTION: Count Word Decompositions into Chemical
# Element Symbols
#
# Examples: Physics has four stated decompositions: P - H
# - Y - Si - C - S P - H - Y - Si - Cs P - H - Y - S - I -
# C - S P - H - Y - S - I - Cs The returned count is 4.
#
# FOLLOW-UPS (answered below, in this order):
# 1. How would you list the decompositions, not just count
# them?
# 2. What if the dictionary is large and symbols can be
# long?
# 3. List the decompositions instead of counting them.
# 4. A larger dictionary with long entries.
# 5. Count modulo a prime, or only test whether any
# decomposition exists.
#
# INTERVIEW NOTES
# Counting the ways to cut a string into dictionary pieces
# is Word Break with a count instead of a yes/no. Define
# ways[i] as the number of decompositions of the suffix
# starting at i; it depends only on ways[i + 1] and ways[i
# + 2], because symbols have one or two letters. Compute
# from the end so every value you need already exists.
#
# BRUTE FORCE (say it first): Recursion that tries a one-
# letter and a two-letter symbol at every position
# At each position take either the next letter or the next
# two letters if they are symbols, and add up the ways to
# finish. Correct and the oracle; the same suffix is
# solved many times. Time exponential (Fibonacci growth).
# Space O(n).


# HELPERS for the brute force
ELEMENTS = set("""h he li be b c n o f ne na mg al si
p s cl ar k ca sc ti v cr mn fe co ni cu zn ga ge as
se br kr rb sr y zr nb mo tc ru rh pd ag cd in sn sb
te i xe cs ba la ce pr nd pm sm eu gd tb dy ho er tm
yb lu hf ta w re os ir pt au hg tl pb bi po at rn fr
ra ac th pa u np pu am cm bk cf es fm md no lr rf db
sg bh hs mt ds rg cn nh fl mc lv ts og""".split())


# BRUTE FORCE (say it first): Recursion that tries a one-
# letter and a two-letter symbol at every position | time
# exponential (Fibonacci growth)
def decompositions_recursive(word, symbols=ELEMENTS):
    w = word.lower()

    def go(i):
        if i == len(w):
            return 1
        total = 0
        if w[i] in symbols:
            total += go(i + 1)
        if i + 2 <= len(w) and w[i : i + 2] in symbols:
            total += go(i + 2)
        return total

    return go(0)


# BASE QUESTION (optimal): Suffix-count DP
# Idea: ways[i] is the number of decompositions of
# word[i:]. A symbol has one or two letters, so ways[i] =
# ways[i + 1] (if word[i] is a symbol) + ways[i + 2] (if
# word[i:i + 2] is a symbol).
def count_decompositions(word, symbols=ELEMENTS,
                         mod=None):
    w, n = word.lower(), len(word)
    # ways[i]: ways to split the suffix starting at i
    ways = [0] * (n + 2)
    ways[n] = 1
    for i in range(n - 1, -1, -1):
        # a one-letter symbol, then a two-letter symbol
        if w[i] in symbols:
            ways[i] += ways[i + 1]
        if i + 2 <= n and w[i:i + 2] in symbols:
            ways[i] += ways[i + 2]
        if mod:
            ways[i] %= mod
    return ways[0]


# FOLLOW-UP 3: List the decompositions instead of counting
# them.
# Answer: Same recursion, but carry the parts chosen so
# far and record them at the end. The number of results
# can be exponential, so ask whether a count or a list is
# wanted.
def list_decompositions(word, symbols=ELEMENTS):
    w, n, memo = word.lower(), len(word), {}

    def splits(i):
        # every way to split the suffix starting at i
        if i == n:
            return [[]]
        if i in memo:
            return memo[i]
        out = []
        for size in (1, 2):
```

```
            part = w[i:i + size]
            if len(part) == size and part in symbols:
                for rest in splits(i + size):
                    out.append([part] + rest)
        memo[i] = out
        return out

    return splits(0)


# FOLLOW-UP 4: A larger dictionary with long entries.
# Answer: Store the dictionary in a set plus the maximum
# word length L (only look at substrings up to L), or in a
# trie, so each position costs O(L) lookups instead of
# O(n).
def count_with_max_len(word, symbols):
    # big dictionary: only look at pieces up to its
    # longest
    longest = max(map(len, symbols))
    w, n = word.lower(), len(word)
    ways = [0] * (n + 1)
    ways[n] = 1
    for i in range(n - 1, -1, -1):
        for size in range(1, min(longest, n - i) + 1):
            if w[i:i + size] in symbols:
                ways[i] += ways[i + size]
    return ways[0]


# FOLLOW-UP 5: Count modulo a prime, or only test whether
# any decomposition exists.
# Answer: Replace addition by addition modulo the prime,
# or by logical or for a yes/no answer. The DP structure
```

```
# is the same.
def exists_decomposition(word, symbols=ELEMENTS):
    # yes / no only: the same DP with or instead of +
    return count_decompositions(word, symbols) > 0


if __name__ == "__main__":
    assert count_decompositions("Physics") == 4
    parts = list_decompositions("Physics")
    assert len(parts) == 4
    assert ["p", "h", "y", "si", "c", "s"] in parts
    assert exists_decomposition("Physics")
    # no element starts with x then q
    assert not exists_decomposition("xq")
    assert count_decompositions("Physics", mod=3) == 1
    assert count_with_max_len("Physics", ELEMENTS) == 4
```

NOTES FROM CANDIDATES

Confirm whether matching is case-sensitive and whether the caller supplies the symbol set or the implementation may treat the periodic table as fixed.
The result counts decompositions, not merely whether at least one decomposition exists.

PREPARATION

Implement a suffix-count or prefix-count formulation and state what each state represents.
Test a word with no decomposition, one decomposition, multiple decompositions, and overlapping one- and two-letter symbols.

STUDY PATH: question 29 of 167 (Core 90)

Priority score 4.05 | Dynamic programming and memoization | no date | frequency Low

MEDIUM | 1Point3Acres | onsite-coding | 7 of 9

Count Same-Color Squares in Matrix

STUDY: TYPE | **ALL TIME 51.2** | **30 DAYS 0** | **PRIORITY 68/228**

ABOUT THIS QUESTION

Topic: Core 90 - Dynamic programming and memoization

Optimal answer from: written and tested here for this exact question

Related in your app (a neighbouring problem, not this exact question): LeetCode #221 Maximal Square

LeetCode match: LC 221 Maximal Square (related)

Frequency: Low | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/count-same-color-squares

QUESTION

LeetCode 221 variant — count maximal same-color square sub-grids. DP on each cell as the bottom-right corner.

Input: m × n grid of colored cells.
Output: count of square sub-grids whose cells are all the same color (or maximum side length, depending on prompt).
DP: dp[i][j] = 1 + min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]) if grid[i][j] == grid[i-1][j] == grid[i][j-1] == grid[i-1][j-1] else 1.
Sum dp[i][j] across the grid for the count variant; max for the side variant.

EXAMPLES

The plain "count all-1 squares" DP generalizes to same-color directly.

APPROACH

Write the all-1-squares DP from memory in under 8 min. Generalize to multi-color in <2 min — only the equality check changes.

HOW TO THINK ABOUT IT

Let dp[i][j] be the side of the largest same-colour square whose bottom-right corner is (i, j). That square exists only if the three neighbours above, left and above-left have the same colour as the cell and can themselves carry a square, so dp is 1 plus the smallest of their dp values. Summing dp over all cells counts every square once (each cell contributes one square per side length up to dp).

Basis: LeetCode 221, Maximal Square, and LeetCode 1277, Count Square Submatrices with All Ones, generalised from 1s to any one colour.

BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
count = 0
for i, j in all_cells(grid):
    for k in range(1, min(m - i, n - j) + 1):
        cells = square(grid, i, j, k)
        if all(c == grid[i][j] for c in cells):
            count += 1
```

TIME	SPACE
O(m * n * min(m, n)^3)	O(1)

WHY IT IS SLOW

Re-checks the same cells again and again, O(m * n * min(m, n)^3). A dp on the bottom-right corner reuses the neighbours' results in O(m * n).

OPTIMAL SOLUTION · dp on the bottom-right corner

TIME	SPACE
O(m * n)	O(n)

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 Return only the largest side instead of the count.

Keep the same dp and take the maximum instead of adding. The common bug is checking only the diagonal neighbour: the up and left neighbours must also match, otherwise a different colour can sit inside the square.

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# MAIN QUESTION: Count Same-Color Squares in Matrix
#
# LeetCode 221 variant — count maximal same-color square
# sub-grids. DP on each cell as the bottom-right corner.
# Input: m × n grid of colored cells. Output: count of
# square sub-grids whose cells are all the same color (or
# maximum side length, depending on prompt). DP: dp[i][j]
# = 1 + min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]) if
# grid[i][j] == grid[i-1][j] == grid[i][j-1] ==
# grid[i-1][j-1] else 1. Sum dp[i][j] across the grid for
# the count variant; max for the side variant. Examples:
# The plain "count all-1 squares" DP generalizes to same-
# color directly.
#
# FOLLOW-UPS (answered below, in this order):
# 1. Return only the largest side instead of the count.
#
# INTERVIEW NOTES
# Let dp[i][j] be the side of the largest same-colour
# square whose bottom-right corner is (i, j). That square
# exists only if the three neighbours above, left and
# above-left have the same colour as the cell and can
```

```
# themselves carry a square, so dp is 1 plus the smallest
# of their dp values. Summing dp over all cells counts
# every square once (each cell contributes one square per
# side length up to dp).
#
# BRUTE FORCE (say it first): Check every square directly
# For every corner and every side length look at all cells
# in the square. Correct; far too slow for a large grid.
# Time O(m * n * min(m, n)^3). Space O(1).


# BRUTE FORCE (say it first): Check every square directly
# | time O(m * n * min(m, n)^3)
def count_squares_brute(g):
    m, n = len(g), len(g[0]) if g else 0
    total = 0
    for i in range(m):
        for j in range(n):
            for k in range(1, min(m - i, n - j) + 1):
                if all(
                    g[a][b] == g[i][j]
                    for a in range(i, i + k)
                    for b in range(j, j + k)
                ):
                    total += 1
    return total


# HELPERS for the base question
def square_sides(g):
    # side[i][j]: the biggest same-colour square whose
    # bottom-right corner is (i, j); one row at a time
    if not g:
        return []
    prev, out = [0] * len(g[0]), []
    for i, row in enumerate(g):
        cur = [1] * len(row)
        for j, c in enumerate(row):
            # all three neighbours must match this cell
            if i and j and g[i - 1][j] == c \
                    and g[i][j - 1] == c \
                    and g[i - 1][j - 1] == c:
                cur[j] = 1 + min(prev[j], cur[j - 1],
                                 prev[j - 1])
        out.append(cur)
        prev = cur
    return out


# BASE QUESTION (optimal): dp on the bottom-right corner
# Idea: dp[j] holds the best square side ending in the
# current row at column j.
def count_same_color_squares(g):
    # every cell contributes one square per side up to its
    # side
    return sum(sum(row) for row in square_sides(g))


# FOLLOW-UP 1: Return only the largest side instead of the
# count.
# Answer: Keep the same dp and take the maximum instead of
# adding. The common bug is checking only the diagonal
# neighbour: the up and left neighbours must also match,
# otherwise a different colour can sit inside the square.
def largest_side(g):
    # the variant that asks for the biggest square only
    return max((max(row) for row in square_sides(g)),
               default=0)
```

```
if __name__ == "__main__":
    # a 3 x 3 block of one colour: 9 + 4 + 1 squares
    block = ["RRR", "RRR", "RRR"]
    assert count_same_color_squares(block) == 14
    assert largest_side(block) == 3
    # a different colour inside blocks the bigger squares
    holes = ["RRR", "RGR", "RRR"]
    assert largest_side(holes) == 1
    assert count_same_color_squares(holes) == 9
    assert count_same_color_squares([]) == 0
    assert largest_side([]) == 0
```

NOTES FROM CANDIDATES

Variant reported in both intern VO and an additional NG round. Common bug: assuming all four corners equal is sufficient — need all three neighbors (top, left, top-left) to match the current cell.

PREPARATION

Write the all-1-squares DP from memory in under 8 min. Generalize to multi-color in <2 min — only the equality check changes.

STUDY PATH: question 30 of 167 (Core 90)

Priority score 3.74 | Dynamic programming and memoization | no date | frequency Single report

MEDIUM | 1Point3Acres | onsite-coding | 8 of 9

Jump Game with Profit

STUDY: TYPE | **ALL TIME 56** | **30 DAYS 0** | **PRIORITY 58/228**

ABOUT THIS QUESTION

Topic: Core 90 - Dynamic programming and memoization

Optimal answer from: written and tested here for this exact question

This question has several parts, so each part has its own answer below.

Related in your app (a neighbouring problem, not this exact question): LeetCode #55 Jump Game

LeetCode match: LC 55 Jump Game (related)

Frequency: Single report | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/jump-game-with-profit

QUESTION

EXAMPLES

Variant-dependent; the principal recurrence is the one above.

APPROACH

Drill the reachability-jump, min-jumps, and coin-change family together so the DP shape is automatic.
Practice clarifying "can I skip without consuming a turn" before coding.

INTERVIEW-DAY PLAN

CLARIFY FIRST

This is underspecified, so ask: is arr[i] the maximum jump (a cap) or an exact jump, is profit collected on landing or when leaving, can you skip a cell for free, are revisits allowed, and what is the goal (reach the end, or maximize profit)?

BRUTE FORCE TO SAY OUT LOUD

Recursively try every jump from every cell. Exponential.

OPTIMAL IDEA

Pick a reading and say it. Cap reading: best[i] is the best profit standing on i; best[i] = profit[i] + max best[j] over j that can reach i; O(n * max jump), or O(n log n) with a heap and lazy deletion. Exact jump with optional skipping: dp[i] = max(dp[i + 1], profit[i] + dp[i + arr[i]]) from the right, O(n). Plain reachability (no profit) is a greedy farthest-reach scan in O(n).

DRY-RUN THESE CASES

A cell with jump 0, an unreachable end, negative profits, a single cell.

LIKELY FOLLOW-UPS

Minimum jumps (LeetCode 45), profits that can be negative, or a limit on the number of jumps.

PITFALLS

INTERVIEW-DAY PLAN

Treating the jump as a cap when it is exact, or the reverse. Not defining what skipping costs.

HOW TO THINK ABOUT IT

This statement is underspecified, so pin it down first (cap or exact jump, when profit is collected, can you skip, what is the goal). The reading with a maximum jump and profit on landing is a DP over positions: the best profit at i is its own profit plus the best of all positions that can reach it. The reading with an exact jump and a free skip is a one-line DP from the right.

Basis: Variant of LeetCode 55 (Jump Game) and LeetCode 45 (Jump Game II). The prompt is underspecified, so two readings are given; confirm which one with the interviewer.

BRUTE FORCE · Reading A: recursion over every jump length

BRUTE FORCE · PSEUDO-CODE

```
best = -INF
def go(i, total):
    global best
    if i == n - 1:
        best = max(best, total)
        return
    for step in range(1, arr[i] + 1):
        if i + step < n:
            gain = profit[i + step]
            go(i + step, total + gain)
```

TIME	SPACE
O(max jump ^ n)	O(n)

WHY IT IS SLOW

Every cell branches into up to max-jump choices, so the paths multiply (max jump to the power n). A table of the best profit per cell, filled left to right, is O(n * max jump).

BRUTE FORCE · Reading B: recursion over 'jump or skip' choices

BRUTE FORCE · PSEUDO-CODE

```
def best(i):
    if i >= n:
        return 0
    skip = best(i + 1)
    jump = 0
    if arr[i] > 0:
        jump = profit[i] + best(i + arr[i])
    return max(skip, jump)
```

TIME	SPACE
O(2^n)	O(n)

WHY IT IS SLOW

BRUTE FORCE · Reading B: recursion over 'jump or skip' choices

Two branches at every cell and the same suffix is solved repeatedly: O(2^n). Computing best(i) once per index from the right is O(n).

OPTIMAL SOLUTION · Reading A: arr[i] is a maximum jump, profit[i] is gained on landing, maximize total profit to the last index

TIME	SPACE
O(n * maxJump)	O(n)

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

OPTIMAL SOLUTION · Reading B: arr[i] is an exact jump; at each index you may jump (gaining profit[i]) or skip one cell for free

TIME	SPACE
O(n)	O(n)

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 Speed up reading A to near-linear time.

The inner loop asks for the max of best[j] over all j that can still reach i. Maintain a max-heap of (best[j], j + arr[j]) and pop entries whose reach is below i (lazy deletion): O(n log n). If the jump limit is the same for all cells, a monotonic deque over a sliding window makes it O(n).

FOLLOW-UP 2 What if you want to know whether the end is reachable at all (plain Jump Game)?

Greedy: scan left to right keeping the farthest reachable index; if i exceeds it you are stuck. O(n) time, O(1) space. Use the DP when profits matter, and the greedy when only reachability matters.

FOLLOW-UP 3 Only reachability, no profit (LeetCode 55).

Greedy: keep the farthest index reachable so far; if the current index is beyond it you are stuck. O(n), O(1).

FOLLOW-UP 4 Make the profit DP faster.

best[i] needs the maximum of best[j] over j that can reach i. A max-heap with lazy deletion of positions that can no longer reach i makes it O(n log n); with a constant maximum jump a monotonic deque gives O(n).

FOLLOW-UP 5 Fewest jumps (LeetCode 45).

BFS by levels or a greedy that tracks the end of the current jump range and the farthest reach, O(n).

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# MAIN QUESTION: Jump Game with Profit
#
# Examples: Variant-dependent; the principal recurrence is
# the one above.
#
# FOLLOW-UPS (answered below, in this order):
# 1. Speed up reading A to near-linear time.
# 2. What if you want to know whether the end is reachable
# at all (plain Jump Game)?
# 3. Only reachability, no profit (LeetCode 55).
# 4. Make the profit DP faster.
# 5. Fewest jumps (LeetCode 45).
#
# INTERVIEW NOTES
# This statement is underspecified, so pin it down first
# (cap or exact jump, when profit is collected, can you
# skip, what is the goal). The reading with a maximum jump
# and profit on landing is a DP over positions: the best
# profit at i is its own profit plus the best of all
# positions that can reach it. The reading with an exact
# jump and a free skip is a one-line DP from the right.
#
# BRUTE FORCE (say it first): Reading A: recursion over
# every jump length
# From each cell try every allowed jump length and add the
# profit of the landing cell; take the best total that
# reaches the last index. Exact and the oracle, and
# exponential. Time O(max jump ^ n). Space O(n).


import heapq


# HELPERS for the brute force
NEG = float("-inf")


# SLOWER VERSION (say it first)
def max_profit_slow(arr, profit):
    # best[i]: most profit when landing on i (arr[j] is
    # the
    # farthest jump from j); O(n^2)
    n = len(arr)
    best = [NEG] * n
    best[0] = profit[0]
```

```
    for i in range(1, n):
        for j in range(i):
            # can j reach i? then landing on i is possible
            if best[j] != NEG and j + arr[j] >= i:
                total = best[j] + profit[i]
                best[i] = max(best[i], total)
    return -1 if best[-1] == NEG else best[-1]


# BASE QUESTION (optimal): Reading A: arr[i] is a maximum
# jump, profit[i] is gained on landing, maximize total
# profit to the last index
# Idea: Let best[i] be the largest profit achievable when
# standing on i. Landing on i from any j < i with j +
```

```python
# arr[j] >= i gives best[i] = profit[i] + max(best[j]).
# Unreachable cells stay at minus infinity.
def max_profit_fast(arr, profit):
    # a max-heap of (-best, reach) with lazy deletion
    n = len(arr)
    best = [NEG] * n
    best[0] = profit[0]
    heap = [(-best[0], arr[0])]
    for i in range(1, n):
        # forget cells that can no longer reach i
        while heap and heap[0][1] < i:
            heapq.heappop(heap)
        if heap:
            best[i] = -heap[0][0] + profit[i]
            heapq.heappush(heap, (-best[i], i + arr[i]))
    return -1 if best[-1] == NEG else best[-1]


# VARIANT: reachability only (LeetCode 55)
# FOLLOW-UP 3: Only reachability, no profit (LeetCode 55).
# Answer: Greedy: keep the farthest index reachable so
# far; if the current index is beyond it you are stuck.
# O(n), O(1).
def can_reach_end(arr):
    # LeetCode 55: only reachability, greedy O(n)
    farthest = 0
    for i, jump in enumerate(arr):
        if i > farthest:
            return False
        farthest = max(farthest, i + jump)
    return True
```

```python
# VARIANT: fewest jumps (LeetCode 45)
# FOLLOW-UP 5: Fewest jumps (LeetCode 45).
# Answer: BFS by levels or a greedy that tracks the end of
# the current jump range and the farthest reach, O(n).
def fewest_jumps(arr):
    # LeetCode 45: levels of a BFS, greedy O(n)
    jumps = end = farthest = 0
    for i in range(len(arr) - 1):
        farthest = max(farthest, i + arr[i])
        if i == end:
            # we must jump now: extend to the farthest
            # reach
            jumps += 1
            end = farthest
    return jumps


if __name__ == "__main__":
    arr, profit = [2, 3, 1, 1, 4], [1, 5, 2, 3, 4]
    # best path: 0 -> 1 -> 2 -> 3 -> 4 = 15
    assert max_profit_slow(arr, profit) == 15
    assert max_profit_fast(arr, profit) == 15
    # the end cannot be reached
    assert max_profit_slow([1, 0, 1], [1, 1, 1]) == -1
    assert max_profit_fast([1, 0, 1], [1, 1, 1]) == -1
    assert can_reach_end(arr)
    assert not can_reach_end([1, 0, 1])
    assert fewest_jumps(arr) == 2
    # the fast version agrees with the slow one
    import itertools
    for jumps in itertools.product(range(4), repeat=5):
        pr = [1, 2, 3, 2, 1]
        assert max_profit_slow(jumps, pr) == (
            max_profit_fast(jumps, pr))
```

NOTES FROM CANDIDATES

Standard intern coding round, single source — clarification with interviewer is key (does "skip" advance by 1 or 0? are revisits allowed?).
Common bug: treating arr[i] as range cap rather than exact jump.

PREPARATION

Drill the reachability-jump, min-jumps, and coin-change family together so the DP shape is automatic.
Practice clarifying "can I skip without consuming a turn" before coding.

STUDY PATH: question 31 of 167 (Core 90)

Priority score 3.61 | Dynamic programming and memoization | no date | frequency Single report

MEDIUM | IN APP LC 354 | 1Point3Acres | phone-screen | 9 of 9

Russian Doll Envelopes

STUDY: TYPE | ALL TIME 52.8 | 30 DAYS 33 | PRIORITY 28/228

ABOUT THIS QUESTION

Topic: Core 90 - Dynamic programming and memoization

Optimal answer from: your app's interview approach (LeetCode #354)

LeetCode match: LC 354 Russian Doll Envelopes

Frequency: Single report | Roles: SWE

Original: https://www.1point3acres.com/interview/problems/company/google/russian-doll-envelopes

QUESTION

LeetCode 354. Maximum chain of envelopes that strictly fit inside each other. Sort + LIS reduction.

Input: list of (width, height) envelopes.
Output: longest chain where each envelope strictly fits in the next.
Trick: sort by width ascending, height descending (when widths tie); run LIS (Longest Increasing Subsequence) on heights.
LIS with patience-sort gives O(n log n).

EXAMPLES

```
[[5,4],[6,4],[6,7],[2,3]] -> 3.
```

APPROACH

Write LIS O(n log n) from memory in under 8 min.
Justify the descending-height-on-tie trick out loud.
Plain LIS and the mountain-LIS variant make good drill partners.

HOW TO THINK ABOUT IT

Sort by width ascending so that only the heights matter, then find the longest strictly increasing subsequence of heights. Envelopes with equal width cannot nest, so sort those by height descending: then a strictly increasing run of heights can never use two envelopes of the same width.

Basis: LeetCode 354, Russian Doll Envelopes.

BRUTE FORCE

BRUTE FORCE · PSEUDO-CODE

```
envelopes.sort()  # by width
best = [1] * len(envelopes)
for i in range(len(envelopes)):
    for j in range(i):
        w1, h1 = envelopes[j]
        w2, h2 = envelopes[i]
        if w1 < w2 and h1 < h2:
            best[i] = max(best[i], best[j] + 1)
answer = max(best)
```

BRUTE FORCE

TIME	SPACE
O(n^2)	O(n)

WHY IT IS SLOW

Each envelope is compared with all earlier ones, O(n^2). After sorting, a longest increasing subsequence of the heights with binary search is O(n log n).

OPTIMAL SOLUTION · Sort, then LIS with binary search

TIME	SPACE
O(n log n)	O(n)

Code: see FULL INTERVIEW CODE below (the optimal solution is its main function).

FOLLOW-UP 1 Why sort heights descending when widths tie?

If two envelopes have the same width they cannot nest, but an ascending height order would let the LIS pick both. Descending order makes a strictly increasing height run use at most one envelope per width.

FULL INTERVIEW CODE · base question + every follow-up

One program for the whole question: the base answer and every follow-up above, built from small helper functions, with a comment above each step. Type it top to bottom in the interview; the demo at the bottom runs it.

```
# MAIN QUESTION: Russian Doll Envelopes
#
# LeetCode 354. Maximum chain of envelopes that strictly
# fit inside each other. Sort + LIS reduction. Input: list
# of (width, height) envelopes. Output: longest chain
# where each envelope strictly fits in the next. Trick:
# sort by width ascending, height descending (when widths
# tie); run LIS (Longest Increasing Subsequence) on
# heights. LIS with patience-sort gives O(n log n).
# Examples: [[5,4],[6,4],[6,7],[2,3]] -> 3.
#
# FOLLOW-UPS (answered below, in this order):
# 1. Why sort heights descending when widths tie?
#
# INTERVIEW NOTES
# Sort by width ascending so that only the heights matter,
# then find the longest strictly increasing subsequence of
# heights. Envelopes with equal width cannot nest, so sort
# those by height descending: then a strictly increasing
# run of heights can never use two envelopes of the same
# width.
#
# BRUTE FORCE (say it first): O(n^2) DP over sorted
# envelopes
# Sort by width and for each envelope take the best chain
# of an earlier envelope that is strictly smaller in both
# dimensions. Correct; quadratic is too slow for 10^5
# envelopes. Time O(n^2). Space O(n).


from bisect import bisect_left
```

```python
# CHECKER: the O(n^2) version
def max_envelopes_slow(envelopes):
    # O(n^2) check used for testing
    ordered = sorted(envelopes)
    best = [1] * len(ordered)
    for i in range(len(ordered)):
        for j in range(i):
            wj, hj = ordered[j]
            wi, hi = ordered[i]
            if wj < wi and hj < hi:
                best[i] = max(best[i], best[j] + 1)
    return max(best, default=0)
```

```python
# BRUTE FORCE (say it first): O(n^2) DP over sorted
# envelopes | time O(n^2)
def doll_brute(envs):
    envs = sorted(envs)
    best = [1] * len(envs)
    for i in range(len(envs)):
        for j in range(i):
            if (
                envs[j][0] < envs[i][0]
                and envs[j][1] < envs[i][1]
            ):
                best[i] = max(best[i], best[j] + 1)
    return max(best, default=0)


# BASE QUESTION (optimal): Sort, then LIS with binary
# search
# Idea: Sort by (width ascending, height descending).
def max_envelopes(envelopes):
    # sort by width; for equal widths put TALLER first so
    # a
    # strictly increasing height run can use only one of
    # them
    ordered = sorted(envelopes,
                     key=lambda e: (e[0], -e[1]))
    # tails[k]: smallest ending height of a chain of k + 1
    tails = []
    for _, h in ordered:
        i = bisect_left(tails, h)
        if i == len(tails):
            tails.append(h)
        else:
            tails[i] = h
    return len(tails)


if __name__ == "__main__":
    envs = [[5, 4], [6, 4], [6, 7], [2, 3]]
    assert max_envelopes(envs) == 3
    assert max_envelopes([]) == 0
    # equal widths cannot nest, however the heights go
    assert max_envelopes([[1, 1], [1, 2], [1, 3]]) == 1
    assert max_envelopes([[4, 5], [4, 6], [6, 7], [2, 3],
```

```python
                          [1, 1]]) == 4
    import itertools
    pts = [(w, h) for w in range(1, 4)
           for h in range(1, 4)]
    for combo in itertools.combinations(pts, 4):
        assert max_envelopes(list(combo)) == (
            max_envelopes_slow(list(combo)))
```

NOTES FROM CANDIDATES

Sorting heights descending on width-ties prevents same-width envelopes from chaining — easy to miss.
LIS via bisect_left on a tails array is the standard O(n log n) implementation.

PREPARATION

Write LIS O(n log n) from memory in under 8 min.
Justify the descending-height-on-tie trick out loud.
Plain LIS and the mountain-LIS variant make good drill partners.